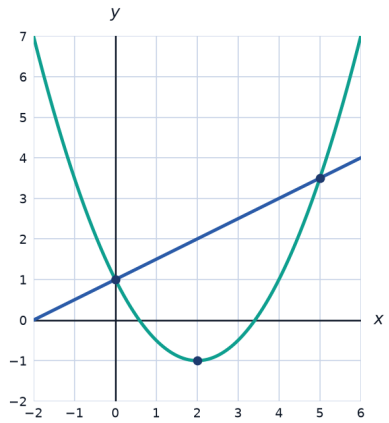
Directed numbers on the number plane

Core Concepts: Directed numbers on the number plane

1 For the model $f(x) = 5x + 6$ in directed numbers on the number plane, evaluate $f(2)$.



a. 16

b. 21

c. 10

d. 20

2 Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

3 A data set recorded for directed numbers on the number plane consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 5x + 6$ in directed numbers on the number plane, evaluate $f(3)$.

a. 21

b. 26

c. 15

d. 25

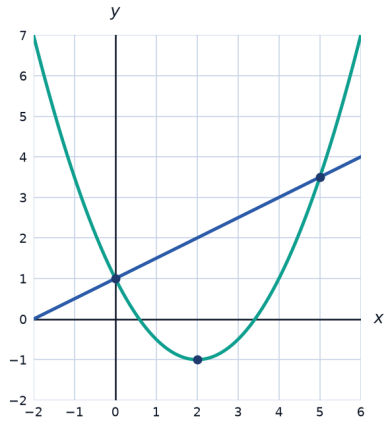
5 Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$



6 A data set recorded for directed numbers on the number plane consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

Graphical & Geometric Interpretation

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7 For the model $f(x) = 5x + 6$ in directed numbers on the number plane, evaluate $f(4)$.

a. 26

b. 31

c. 20

d. 30

8 Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

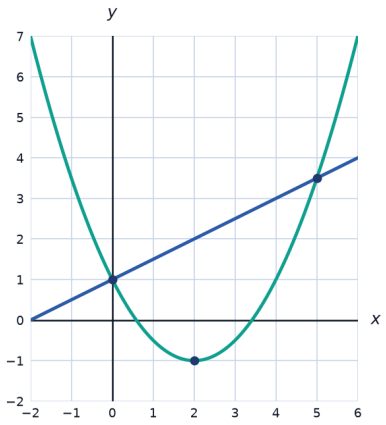
9 A data set recorded for directed numbers on the number plane consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

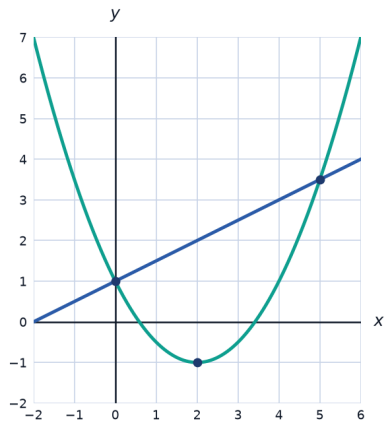
d. 18.5



Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 5x + 6$ in directed numbers on the number plane, evaluate $f(5)$.



a. 31

b. 36

c. 25

d. 35

11 Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

12 A data set recorded for directed numbers on the number plane consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

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